(a) Because the last component of h is the second to last square, and no matter what you roll, 1 to N, you will always move to the last space, so its guaranteed to be 1. as in 1 move to get to the end of the space.

```
% Enter the SL vector below
SL = [24 2; 4 22; 20 6; 8 18; 16 10; 12 14; 15 11; 23 3; 5 21];
```

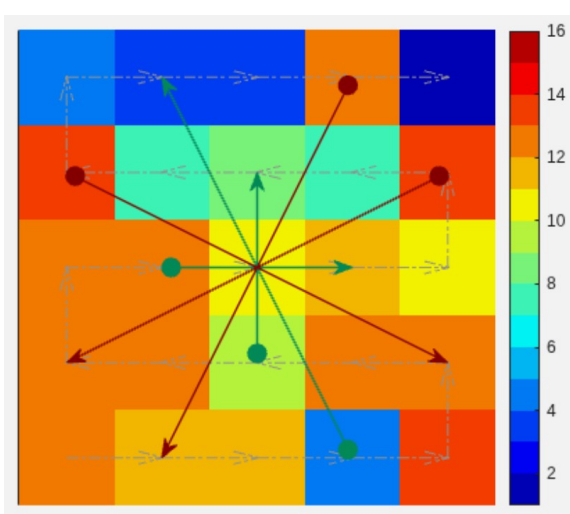
(b) This is approaching
game theory than math, but originally I had made a small lua script to create the small sl vector.

```lua
local function rndm(str, x, y, neg)
    if x > y / 2 then
        return str
    end
    local onenum
    local twonum
    onenum = y + 1 - x
    twonum = x
    if neg < 0 then
        twonum = onenum
        onenum = x
    end
    return rndm(str .. onenum .. " " .. twonum .. "; ", x + 2, y, -neg)
end
print(rndm("", 2, 25, 1))
```

It returned the following output:

```
24 2; 4 22; 20 6; 8 18; 16 10; 12 14;
```

this is six snakes and ladders, I like that the last space is instant death to the shadow realm, it adds a spicyness, like you can do everything right and still fail. And being able to win early is good. I also like how the snakes are overall worse to get hit by then the ladders overall. I think red is good for this game.




```
h =

   12.1660
   11.7870
   11.8963
    4.6530
   13.0656
   12.9500
   12.6439
    9.5134
   12.5519
   12.7185
   12.0160
   12.2562
   10.8072
   11.2562
   10.2570
   13.7185
    7.8011
    8.5134
    7.2972
   13.9500
    4.2619
    3.6530
    3.1312
   12.7870
```

(c) 12.1160, its the first value of h, corresponding to how many moves is likely to happen at first space.
(d)

```
for i= 1:height(SL)
   s(SL(i,1)) = 0;
end
```

Its 3 lines of code, just add it after s, no problems at all.

add this line after. The way to understand this is by first looking at the recursive formula. Which is h = s + ph. The s just adds after moving to a space, in the case theoretically if you made one of the numbers 0 in the s index vector, it would just not add the extra move as a move. This is how the snake works.

Then we can move it back into our real formula for h, with the understanding out of the way.

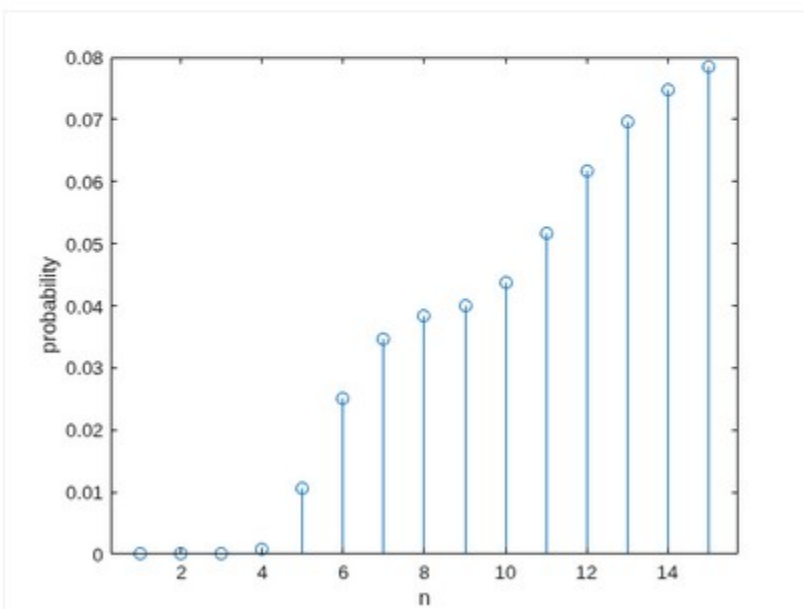
PART 2

(a) done

(b) code below next page, output on the right.

```
SL =

    18     1
    20     2
    21     3
    22     4
    23     5
    24     6
```



```
% We build the matrix P and you are required to use P to calculate the probability of winning by the
% end of turn n. Enter your solution in Ans below. Note that sol records the answer for different values of
% n and is then used to plot the graph

% Enter SL below
SL = [18 1; 20 2; 21 3; 22 4; 23 5; 24 6]

% Setting up P

% Construct P, ignorning the snakes and ladders
P = zeros(25);
for i=1:25
    for j=i+1:min(i+6,25)
        P(i,j) = 1/6;
    end
end
for i = 1:6
    P(25-6+i, 25) = min((i+1)/6, 1);
end

% Adjust for the snakes and ladders
no_SL=height(SL);
for i= 1:no_SL
    for j=1:25
        if j == SL(i,2)
            P(SL(i,1),j) = 1;
        else
            P(SL(i,1),j) = 0;
        end
    end
end

% Record the probability for different values of n
sol = [];
for n=1:10

    % Your value for the probability of winning by the end of turn n goes below
end
for n=1:15
    Rndm = P^n;
    Ans = Rndm(1, 25);;

    sol = [sol, Ans];
end

% Plot it out
stem(sol);
xlabel('n')
ylabel('probability')
```